

Introductory Exercises on Relational Databases

CCMC – NAHPI, UBa

Context

You are hired by a small university to help design and understand its **Student Course Registration System (SCRS)**. The university wants to manage information about students, courses, and the instructors who teach them.

You will gradually design, analyze, and reason about this system in the exercises below.

1. Exercise 1 – Identify Entities and Relationships

Objective: Understand and extract entities, attributes, and relationships from a description.

Task: Read the following description carefully:

Each *student* has a unique student ID, a name, a birth date, and a study program. Each *course* has a course code, a title, and a number of credits. Each *instructor* has an instructor ID, name, and department. An instructor can teach several courses, but each course is taught by only one instructor. A student can register for many courses, and a course can have many students.

- a) Identify all **entities** and their **attributes**.
- b) Identify all **relationships**, their **types** (1–1, 1–N, or N–M), and **participation constraints**.
- c) Suggest **primary keys** for each entity.

2. Exercise 2 – Create the Relational Schema

Objective: Convert a conceptual design to a relational schema.

Task: Based on your answer to Exercise 1, design a **relational schema** for the Student Course Registration System.

Include:

- Table names
- Column names with data types (use generic SQL types: INT, VARCHAR, DATE, etc.)
- Primary keys, foreign keys, and referential integrity constraints

Hint: Think carefully about how to represent the many-to-many relationship between Students and Courses.

3. Exercise 3 – Analyze Functional Dependencies

Objective: Understand normalization and data redundancy.

Task: Consider the following relation:

Enrollment(StudentID, StudentName, CourseCode, CourseTitle,
InstructorName)

- a) Identify all **functional dependencies** that might exist in this relation.
- b) Explain what **anomalies** could occur (insertion, deletion, update).
- c) Propose a **normalized decomposition** (up to 3NF) of the relation and justify it.

4. Exercise 4 – Reasoning About Keys and Constraints

Objective: Apply reasoning to ensure data integrity.

- a) For your schema from Exercise 2, identify all **candidate keys** (if any) in each table.
- b) Specify all **foreign key constraints** that ensure referential integrity.
- c) Give one **example of a violation** of referential integrity and explain how the DBMS would prevent it.

5. Exercise 5 – Writing and Interpreting Relational Queries

Objective: Test understanding of relational data and logical query formulation.

Task: Given your schema, formulate SQL-style relational queries for:

- 1) List all students enrolled in “Database Systems.”
- 2) Find the names of instructors who teach more than one course.
- 3) Find the total number of students registered per course.
- 4) Find students who are not registered in any course.
- 5) Suppose the course “Introduction to Databases” is deleted. Explain what happens to related records in your schema and how you would enforce data consistency using referential actions (e.g., ON DELETE CASCADE, ON DELETE SET NULL, etc.).